

KATHMANDU UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF CIVIL ENGINEERING



LABORATORY REPORT
OF
REINFORCED CONCRETE STRUCTURE DESIGN (CIEG 310)

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1 DETERMINATION OF THE FLEXURE STRENGTH OF BEAM IN VARIOUS CASES

2 Title

- i. Determination of Flexural Strength of Plain Concrete
- ii. Determination of Ultimate Load and Load versus deflection of Molded Over Reinforced Beam (Flexure Specimen)
- iii. Determination of Ultimate Load and Load versus deflection of Molded Under Reinforced Beam (Flexure Specimen)
- iv. Determination of Ultimate Load and Load versus deflection of Molded Under Reinforced Beam with Shear Stirrups (Flexure and Shear Specimen)

3 Objective

- i. To determine the Ultimate Load
- ii. To determine the Flexure Strength of Beam
- iii. To observe the cracking patterns in and concrete beam
- iv. To determine the Load Vs. Deflection for each case
- v. To compare Load Vs Deflection graphs among all

4 Theory

A beam, as a flexure member, is primarily designed to resist bending forces induced by loads applied perpendicular to its longitudinal axis. In structural engineering, beams play a crucial role in transferring loads from slabs, floors, and roofs to columns and foundations. The performance of a beam under flexure depends on its material properties, cross-sectional shape, and the type and amount of reinforcement used.

4.1 Flexural Behavior of Beams

When a beam is subjected to transverse loads, it experiences bending moments that induce tensile and compressive stresses across its cross-section. The top fibers of the beam typically undergo compression, while the bottom fibers experience tension. The neutral axis, where the stress is zero, divides these two regions.

4.2 Types of Failure in Beams

- i. **Under-Reinforced Beam**

An under-reinforced beam has less reinforcement than the amount required to balance the section. This type of beam will fail by yielding steel reinforcement before the concrete reaches its ultimate compressive strength. This is a ductile failure mode, meaning the beam will show significant deflection and cracking before failure, providing ample warning signs. The tensile steel yields, and the beam exhibits large deformations and extensive cracking. This ductility is desirable in structural design because it ensures that failure is gradual and not sudden.

ii. No Reinforcement (Plain Concrete Beam)

A beam without any reinforcement relies entirely on the tensile strength of concrete. Concrete is very weak in tension compared to compression. A plain concrete beam will fail suddenly and in a brittle manner when the tensile stress exceeds the tensile strength of the concrete. The beam cracks and fails abruptly with little warning. This type of failure is dangerous and undesirable in structural applications.

iii. Balanced Section

A balanced section is designed such that the steel reinforcement yields at the same time the concrete reaches its ultimate compressive strength. In theory, both steel and concrete reach their failure points simultaneously. This is an idealized condition where the beam exhibits both ductile and brittle characteristics. The beam would show moderate warning signs before failure. This type of design ensures an optimized use of materials and balanced performance.

iv. Over-Reinforced Beam

An over-reinforced beam contains more reinforcement than the amount required for a balanced section. The concrete will fail in compression before the steel reinforcement yields. This leads to a brittle failure mode. The beam will fail suddenly and catastrophically without significant prior deformation or cracking. This lack of ductility makes over-reinforced beams less desirable because they do not provide adequate warning before failure.

4.3 Shear Failure Pattern of Beams

Shear failure in beams occurs when the shear stresses exceed the shear strength of the material. Unlike bending failure, which involves the material's tensile and compressive strengths, shear failure is characterized by the material sliding along a plane parallel to the applied force. Understanding the different types of shear cracks and shear failure modes is crucial for designing beams that can safely carry loads without catastrophic failure.

4.4 Types of Shear Cracks

i. Diagonal Tension Cracks

These cracks form at approximately 45 degrees to the longitudinal axis of the beam. They typically initiate near the support where the shear forces are highest and propagate toward the mid-span of the beam. They appear as diagonal lines extending from the point of maximum shear stress, often starting near the support and moving upward toward the top of the beam.

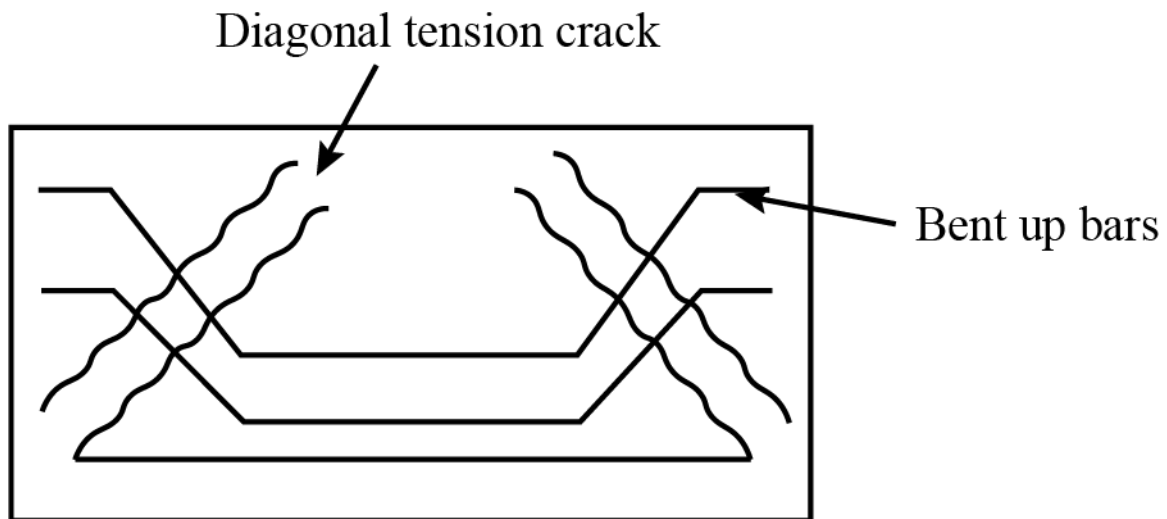


Figure 1. Diagonal Tension Crack

ii. Flexure-Shear Cracks

These cracks are a combination of flexural and shear stresses. They usually start as flexural cracks in the tension zone of the beam and then propagate diagonally due to the presence of shear stresses. These cracks often start vertically in the tension zone and then curve or branch out diagonally due to the combined action of bending and shear forces.

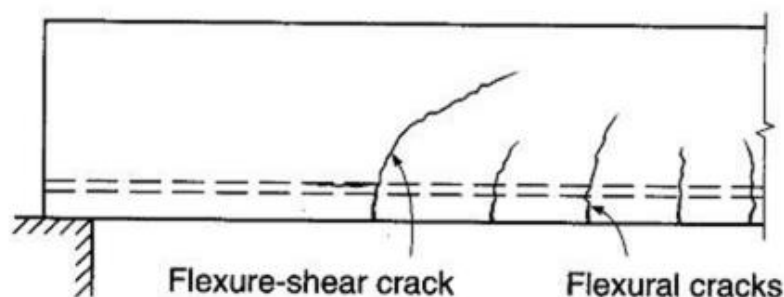


Figure 2. Flexure-Shear Crack

iii. Web Shear Cracks

These cracks occur in the web of the beam without the presence of significant flexural cracking. They form directly due to high shear forces in the web region. These cracks appear as inclined cracks starting from the neutral axis of the beam and propagating toward the compression or tension zones.

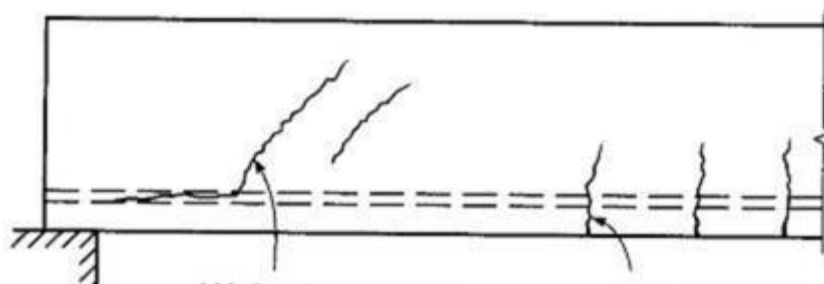


Figure 3. Web Shear Crack

4.5 Major Types of Shear Failure

i. Diagonal Tension Failure

This type of failure occurs due to the formation and propagation of diagonal tension cracks. When the tensile stresses in the diagonal direction exceed the tensile strength of the concrete, diagonal tension cracks form and propagate rapidly, leading to sudden and brittle failure. The beam exhibits sudden failure with little warning, characterized by a significant diagonal crack extending from the support to the mid-span.



Figure 4. Diagonal Tension Failure

ii. Shear Compression Failure

This type of failure occurs when the compression zone of the beam is crushed due to high shear forces. When the diagonal compression stresses in the web of the beam exceed the compressive strength of the concrete, it leads to crushing in the compression zone. The beam shows crushing and buckling in the compression zone near the support, often accompanied by web shear cracks.

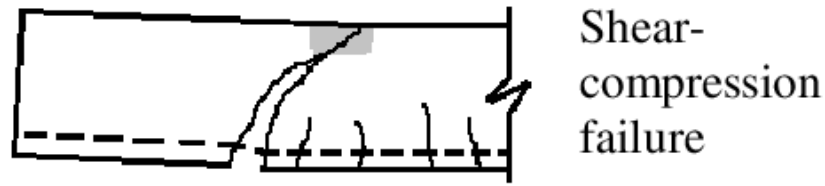


Figure 5. Shear-Compression Failure

iii. Shear-Flexure Failure

This failure mode is a combination of flexural and shear failures. The beam initially develops flexural cracks, which then propagate and transform into shear cracks due to high shear forces. This combined effect leads to failure. The beam exhibits both flexural and shear cracks, with a failure pattern that includes significant deformation and cracking in both directions.



Figure 6. Shear-Flexure Failure

iv. Web Crushing Failure

This occurs in beams with thin webs subjected to high shear forces. When the web of the beam cannot carry the shear forces, it leads to crushing and splitting of the web. The failure is characterized by the crushing of the concrete in the web, leading to a sudden and brittle collapse of the beam.

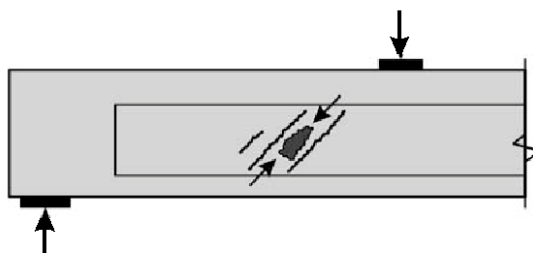


Figure 7. Web Crushing Failure

4.6 Compressive Strength of Concrete

The compressive strength of concrete is a crucial parameter that defines its capacity to withstand axial compressive loads without failure. This property is typically assessed through

a standardized test involving concrete cubes, which are cast and cured under controlled conditions to ensure uniform hydration and strength development. After the curing period, these cubes are placed in a compression testing machine, where a gradually increasing load is applied until the specimen fails. The maximum load at which failure occurs, denoted as P , is recorded. The compressive strength f_c is then calculated using the formula:

$$f_c = \frac{P}{A}$$

where A is the cross-sectional area of the cube. The result is expressed in megapascals (MPa). This calculated value is essential for verifying that the concrete meets the specific design criteria and construction standards. Ensuring adequate compressive strength is vital for the structural integrity, safety, and durability of concrete structures, as it directly affects their ability to bear loads, resist various stressors, and maintain their functional performance over their intended lifespan.

4.7 Relation among Compressive Strength and Flexural Strength

According to the Indian Standards IS 516 and IS 456, there is a recognized relationship between the compressive strength and flexural strength of concrete. While compressive strength measures the ability of concrete to withstand axial loads, flexural strength assesses its capacity to resist bending or flexural stresses. IS 456 provides empirical formulas to estimate the flexural strength from the compressive strength. For ordinary concrete, the flexural strength f_{cr} can be estimated using the following formula:

$$f_{cr} = 0.7\sqrt{f_c}$$

where f_{cr} is the flexural strength in MPa and f_c is the compressive strength in MPa. This relation indicates that as the compressive strength of concrete increases, its flexural strength also increases, but at a lower rate. The empirical formula is useful in preliminary design and quality control, allowing engineers to estimate flexural strength based on more commonly tested compressive strength values. This relationship underscores the inherent connection between the two properties, both of which are critical for ensuring the structural performance and durability of concrete elements in various applications.

5 Methodology

5.1 Material Collection

Firstly, the quantities of materials needed for the concrete mix were determined, and the appropriate amounts of cement, sand, and aggregate (both fine and coarse) were collected for mixing. The following materials were used for lab test of reinforced concrete:

- i. Cement
- ii. Fine aggregate (<4.75mm)
- iii. Coarse aggregate (> 4.75mm)
- iv. Iron Bar
- v. Mold

The sand was sieved using a sieve of 4.75mm. The retained coarse aggregate and the fine aggregate that passed through the sieve were collected separately. The cement and iron bar were available in the laboratory.

5.2 Material Preparation

All materials shall be brought to room temperature, preferably $27^{\circ}\pm 3^{\circ}\text{C}$ before commencing the tests.

The cement samples, on arrival at the laboratory shall be thoroughly mixed dry either by hand or in a suitable mixer in such a manner as to ensure the greatest possible blending and uniformity in the material, care being taken to avoid the intrusion of foreign matter. The cement shall then be stored in a dry place, preferably in air-tight metal containers.

Samples of aggregates for each batch of concrete shall be of the desired grading and shall be in an air-dried condition. In general, the aggregate shall be separated into fine and coarse fractions and recombined for each concrete batch in such a manner as to produce the desired grading. IS Sieve 480 shall be normally used for separating the fine and coarse fractions, but where special grading is being investigated, both fine and coarse fractions shall be further separated into different sizes.

5.3 Mixing Concrete

The concrete batch was mixed on a water-tight, non-absorbent platform with a shovel, trowel or similar suitable implement, using the following procedure:

- a. The cement and fine aggregate were mixed dry until the mixture is thoroughly blended and is uniform in color,
- b. The coarse aggregate was then added and mixed with the cement and fine aggregate until the coarse aggregate is uniformly distributed throughout the batch, and
- c. The water was then added, and the entire batch was mixed until the concrete appears to be homogeneous and has the desired consistency.



Figure 8. Dry Mixing of cement, sand and aggregate

5.4 Size of Specimens

The standard size shall be 15× 15× 70 cm. Alternatively, if the largest nominal size of the aggregate does not exceed 19mm, specimens 10 × 10 × 50 cm may be used.



Figure 9. Mould

5.5 Rebar Cutting

In this process, an iron bar was cut to the required length.

- i. An iron bar of required length was measured with measuring tape and marked at intervals of 66 cm.
- ii. Then it was clamped to prevent moving during cutting and ensure straight cut.

- iii. The iron bar was cut at the marked points using a circular handheld cutter. A total of 5 rebars each of 66 cm were cut. (3 number of rebar for over-reinforced and 2 number of rebar for under-reinforced)

5.6 Rebar Binding

Binding wires of required length were cut using pliers. The rebar was placed in a designated position. A piece of binding wire was taken and placed around the intersection of the two rebars. The wires were twisted tightly to ensure they can hold rebar curing concrete placing.

5.7 Rebar Placing

Over-reinforced: 3 Nos. of 8 mm diameter rebar were placed maintaining the 2cm clear cover from the bottom /tensile face of the concrete.

Under-reinforced: 2 Nos. of 8 mm diameter rebar were placed maintaining the 2cm clear cover from the bottom /tensile face of the concrete.

Shear Stirrups: Rebar of 500Mpa of diameter & length was used as shear reinforcement as shown in the figure below according to calculation done.



Figure 10. Placing of rebar

5.8 Tamping Bar

The mould was tamped with a steel bar weighing 2 kg, 40 cm long with a ramming face 25mm square.

5.9 Concreting

The mould was of metal, preferably steel or cast iron and the metal was of sufficient thickness to prevent spreading or warping. The mould was constructed with the longer dimension horizontal in such a manner as to facilitate the removal of the moulded specimens without damage. The freshly mixed concrete was placed into a $15 \times 15 \times 70$ cm mould. The concrete mix was placed in layers, to ensure uniform distribution. After placing each layer, it was compacted with the help of a tamping rod to remove any entrapped air which could weaken the specimen. Ensuring proper compaction is essential for achieving the desired density and strength. After the mould is filled and compacted, the surface is leveled and smoothed with a trowel. The filled mould is then covered to prevent moisture loss and kept in a location with controlled temperature and humidity for curing.



Figure 11. Concreting and Placement in the mould

5.10 Curing

The test specimens were stored in a curing tank at a temperature of $27 \pm 2^\circ\text{C}$ for 28 days from the time of addition of water to the dry ingredients. The specimens shall not be allowed to dry at any time until they have been tested.

5.11 Testing

5.11.1 Apparatus

The testing machine may be of any reliable type of sufficient capacity for the tests and capable of applying the load at any suitable rate. The permissible errors shall be not greater than ± 0.5 percent of the applied load where a high degree of accuracy is required and not greater than $\pm$

1.5 percent of the applied load for commercial type of use. The bed of the testing machine shall be provided with two steel rollers, 38mm in diameter, on which the specimen is supported, and these rollers shall be so mounted that the distance from the center to center is 60 cm for 15.0 cm specimens or 40 cm for 10.0 cm specimens. The loads shall be applied through two similar rollers mounted at the third points of the supporting span, that is spaced at 20 or 13.3 cm center to center. The load shall be mounted in such a manner that the load is applied axially and without subjecting the specimen to any torsional stress or restraints One suitable arrangement which complies with these requirements is indicated in **Figure 12** and the machine is shown in **Figure 13**.

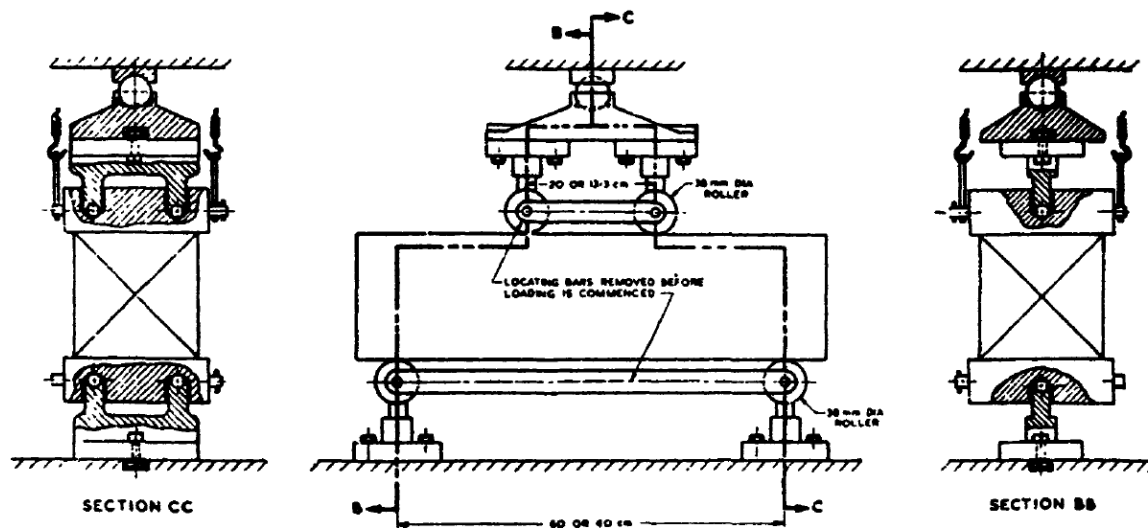


Figure 12. Arrangement for loading of Flexure test specimen



Figure 13. Testing Machine

5.11.2 Procedure

Test specimens stored in water at a temperature of 24° to 30°C for 14days before testing, was tested immediately on removal from the water whilst they are still in a wet condition. The dimensions of each specimen were noted before testing. No preparation of the surfaces is required.

5.11.2.1 *Placing the Specimen in the Testing Machine*

The bearing surfaces of the supporting and loading rollers was wiped clean, and any loose sand or other material or other material was removed from the surfaces of the specimen where they are to contact the rollers. The specimen was placed in the machine in such a manner that the load shall be applied to the uppermost surface as cast in the mould, along two lines spaced 20.0 or 13.3 cm apart. The axis of the specimen was carefully aligned with the axis of the loading device. No packing was used between the bearing surfaces of the specimen and the rollers. The load was applied without shock and increasing continuously at a suitable rate. The load was increased until the specimen failed, and the maximum load applied to the specimen during the test was recorded. The appearance of the fractured faces of concrete and any unusual features in the type of failure was noted and shown in Figure 18,19,20 and 21.

6 Observations

6.1 For Plain Beam, Cube and Under-reinforced beam with shear stirrups

6.1.1 Material Calculation for Mixing

i. For compressive strength of cube:

Size of cube specimen: 10cm*10cm*10cm

Number of cube specimen: 3

Total volume of cube specimen (V_1) = $3 * (10 * 10 * 10) \text{ cm}^3$

$$= 3000 \text{ cm}^3$$

$$= 3 * 10^{-3} \text{ m}^3$$

ii. For Plain Beam and under-reinforced beam with shear stirrups:

Number of Specimen: 2

Total Volume of specimen (V_2) = $2 * (50 * 10 * 10) \text{ cm}^3$

$$= 10^4 \text{ cm}^3$$

$$= 0.01 \text{ m}^3$$

Therefore, Total volume of concrete required = Total volume of cube specimen (V_1)

+ Total volume of specimen (V_2)

$$= 3 * 10^{-3} \text{ m}^3 + 0.01 \text{ m}^3$$

$$= 0.013 \text{ m}^3$$

Now,

Total Dry volume of concrete required = $1.5 * 0.013 \text{ m}^3$

$$= 0.019 \text{ m}^3$$

Since, our concrete is of M20 grade so ratio of Cement, Sand and Aggregate

(C:S: A) = 1:1.5:3

a. Quantity of Cement = $1/5.5 * 0.019 \text{ m}^3$

$$= 3.45 * 10^{-3} \text{ m}^3$$

Mass of Cement = Volume * Density

$$= 3.45 * 10^{-3} \text{ m}^3 * 1440 \text{ kg/m}^3 \text{ (For OPC cement, taking density as } 1440 \text{ kg/cm}^3\text{)}$$

$$= 4.96 \text{ kg}$$

$$= 5 \text{ kg (Approx)}$$

$$\text{b. Quantity of Sand} = 1.5/5.5 * 0.019 \text{ m}^3$$

$$= 5.18 * 10^{-3} \text{ m}^3$$

$$\text{Mass of Sand} = \text{Volume} * \text{Density}$$

$$= 5.18 * 10^{-3} \text{ m}^3 * 1600 \text{ kg/m}^3 \text{ (For sand, taking density as } 1600 \text{ kg/m}^3\text{)}$$

$$= 8.28 \text{ kg}$$

$$= 9 \text{ kg (Approx)}$$

$$\text{c. Quantity of Aggregate} = 3/5.5 * 0.019 \text{ m}^3$$

$$= 0.01 \text{ m}^3$$

$$\text{Mass of Aggregate} = \text{Volume} * \text{Density}$$

$$= 0.01 \text{ m}^3 * 1550 \text{ kg/m}^3$$

$$= 16 \text{ kg}$$

Now,

We know, the water to cement ratio (W:C) = 0.5

$$\text{Quantity of water} = 0.5 * 4.96 \text{ kg}$$

$$= 2.48 \text{ liters}$$

$$= 2.5 \text{ liters}$$

Table 1. Flexural Strength Calculation of Plain Beam

Flexural Strength Testing of Molded Flexure Test Specimens IS 516:1959/2021										
Location:		Concrete Technology Lab, KU			Brand of Cement:		Sarbottam			
Date of Casting:		14/05/2024			Aggregate Detail:		20mm passing & 10 mm retaining			
Date of Testing:		11/06/2024			Concrete Grade:		M20			
Age of Beam:		28 Days			W/C Ratio:		0.5			
Curing Condition:		Water Curing			ID Mark:					
S.N.	Beam No.	Dimension of Beam (mm)	Weight of Specimen (Kg.)	Position of Fracture (mm)	Length of Beam (mm)	Failure Load (N)	Flexural Strength (N/mm ²)	Average Strength (N/mm ²)	Failure Mode	Rem .
1	Plain Beam	500*100*100	11.01	145	400	10320	4.048		Flexure	28 days

6.1.2 Load vs Deflection Curve of Plain Beam

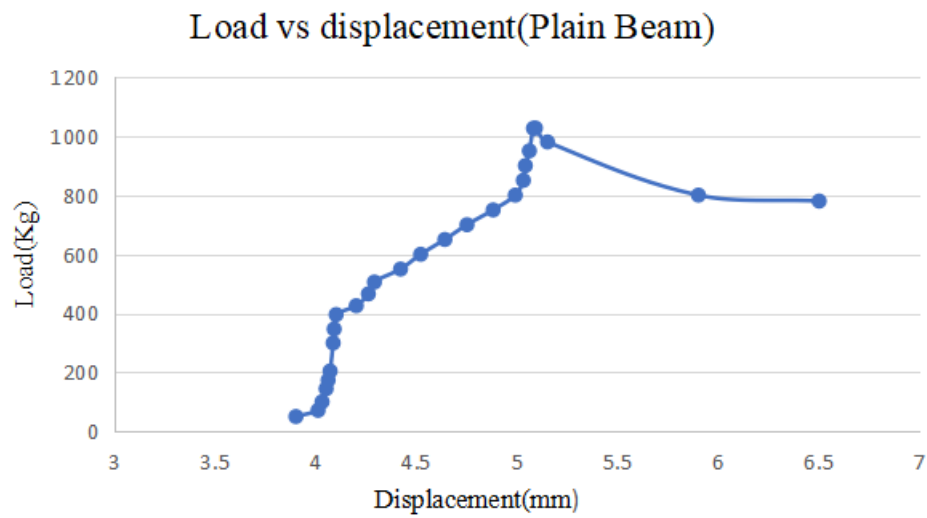


Figure 14. Load vs Deflection curve of Plain Beam

From graph, the ultimate load at the time of failure is 1027 kg.

6.2 For Over-Reinforced

6.2.1 Material Calculation for Mixing

Inside length of the mold = 700 mm

Inside breadth of mold = 150 mm

Inside depth of mold = 150 mm

∴ Volume of 2 beams = $2 \times 700 \times 150 \times 150 \text{ mm}^3 = 0.0315 \text{ m}^3$

Total volume required = 0.0315 m^3

The volume of concrete was increased by 54 % to obtain dry volume.

Hence, dry volume of concrete = 0.04851 m^3

Density of cement = 1440 kg/m^3

Density of sand = 1600 kg/m^3

Density of coarse aggregate = 1800 kg/m^3

Total part of the mixture = $1 + 1.5 + 3 = 5.5$

Weight of cement = $1/5.5 \times (0.04851) \times 1440 = 12.7008 \text{ kg}$

Weight of sand = $1.5/5.5 \times (0.04851) \times 1600 = 21.168 \text{ kg}$

Weight of coarse aggregate = $3/5.5 \times (0.04851) \times 1800 = 47.628 \text{ kg}$

Volume of water required = $0.4 \times 12.7008 = 5.08 \text{ L}$

6.2.2 Check of UR, OR or Balanced section

We have, Compressive Strength of Steel (f_y) = 500 MPa

Compressive Strength of Concrete (f_{ck}) = 20 MPa

Limiting depth of Neutral Axis (x_m) = $0.46 \times (150 - 10 - \Phi_{tie} - \Phi_l/2)$

$$= 0.46 \times (150 - 10 - 6 - 10/2) = 59.34 \text{ mm}$$

Assuming 3 Nos. of 8 mm dia. Bars,

$$\text{Depth of Neutral Axis (x)} = \frac{0.87 \times f_y \times A_{st}}{0.36 \times f_{ck} \times b} = \frac{0.87 \times 500 \times 3 \times \pi \times 8^2}{4 \times 0.36 \times 20 \times 150} = 60.73 \text{ mm}$$

As $x > x_m$, the section is Over-Reinforced.

Hence, adopt 3 Nos. of 8 mm dia. Bars.

Table 2. Flexural Strength Calculation of Over-Reinforced Beam

Flexural Strength Testing of Molded Flexure Test Specimens IS 516:1959/2021										
Location:		Concrete Technology Lab, KU			Brand of Cement:		Sarbottam			
Date of Casting:		14/05/2024			Aggregate Detail:		20mm passing & 10 mm retaining			
Date of Testing:		11/06/2024			Concrete Grade:		M20			
Age of Beam:		28 Days			W/C Ratio:		0.4			
Curing Condition:		Water Curing			ID Mark:					
S.N.	Beam No.	Dimension of Beam (mm)	Weight of Specimen (Kg.)	Position of Fracture (mm)	Length of Beam (mm)	Failure Load (N)	Flexural Strength (N/mm ²)	Average Strength (N/mm ²)	Failure Mode	Rem .
1	OR	700*150*150	39.4	1(From Support)	600	88270	0.078		Shear	28days

6.2.3 Load vs Deflection curve of Over-reinforced Beam

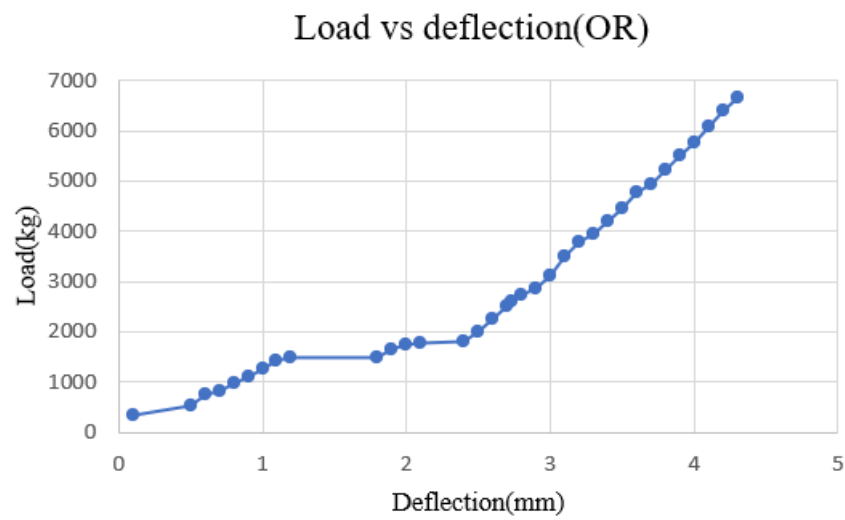


Figure 15. Load vs Deflection Curve of Over-reinforced Beam

From graph, the ultimate load at the time of failure is 882kg.

6.3 For Under-reinforced:

6.3.1 Material Calculation for Mixing

Inside length of the mold = 700 mm

Inside breadth of mold = 150 mm

Inside depth of mold = 150 mm

∴ Volume of 2 beams = $2 \times 700 \times 150 \times 150 \text{ mm}^3 = 0.0315 \text{ m}^3$

Total volume required = 0.0315 m^3

The volume of concrete is increased by 54 % to obtain dry volume.

Hence, dry volume of concrete = 0.04851 m^3

Density of cement = 1440 kg/m^3

Density of sand = 1600 kg/m^3

Density of coarse aggregate = 1800 kg/m^3

Total part of the mixture = $1 + 1.5 + 3 = 5.5$

Weight of cement = $1/5.5 \times (0.04851) \times 1440 = 12.7008 \text{ kg}$

Weight of sand = $1.5/5.5 \times (0.04851) \times 1600 = 21.168 \text{ kg}$

Weight of coarse aggregate = $3/5.5 \times (0.04851) \times 1800 = 47.628 \text{ kg}$

Volume of water required = $0.4 \times 12.7008 = 5.08 \text{ L}$

6.3.2 Check of UR, OR or Balanced section

We have, Compressive Strength of Steel (f_y) = 500 MPa

Compressive Strength of Concrete (f_{ck}) = 20 MPa

Limiting depth of Neutral Axis (x_m) = $0.46 \times (150 - 10 - \Phi_{tie} - \Phi_l/2)$ [From IS 456:2000]

$$= 0.46 \times (150 - 10 - 6 - 10/2) = 59.34 \text{ mm}$$

Assuming 2 Nos. of 8 mm dia. Bars,

$$\text{Depth of Neutral Axis (x)} = \frac{0.87 \times f_y \times A_{st}}{0.36 \times f_{ck} \times b} = \frac{0.87 \times 500 \times 2 \times \pi \times 8 \times 8}{4 \times 0.36 \times 20 \times 150} = 40.49 \text{ mm}$$

As $x < x_m$, the section is Under-Reinforced.

Hence, adopt 2 Nos. of 8 mm dia. Bars.

Table 3. Flexural Strength Calculation of Under-Reinforced Beam

Flexural Strength Testing of Molded Flexure Test Specimens IS 516:1959/2021										
Location:		Concrete Technology Lab, KU			Brand of Cement:		Sarbottam			
Date of Casting:		14/05/2024			Aggregate Detail:		20mm passing & 10 mm retaining			
Date of Testing:		11/06/2024			Concrete Grade:		M20			
Age of Beam:		28 Days			W/C Ratio:		0.4			
Curing Condition:		Water Curing			ID Mark:					
S.N.	Beam No.	Dimension of Beam (mm)	Weight of Specimen (Kg.)	Position of Fracture (mm)	Length of Beam (mm)	Failure Load (N)	Flexural Strength (N/mm ²)	Average Strength (N/mm ²)	Failure Mode	Rem .
1	UR	700*150*150	38.54	5(from Support)	600	55200	0.245		Shear	28days

6.3.3 Load vs Deflection Curve of Under-reinforced beam

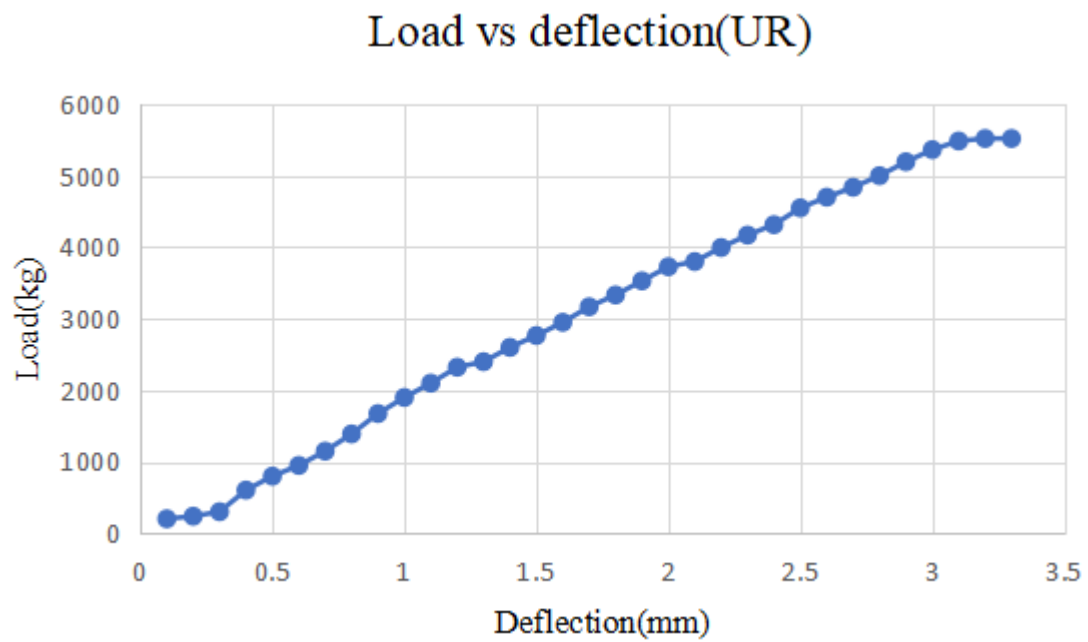


Figure 16. Load vs Deflection Curve of Under-reinforced beam

From graph, the ultimate load at the time of failure is 5520 kg.

6.4 For Under-Reinforced with Shear Stirrups:

6.4.1 Check of UR, OR or Balanced Section

For 6mm diameter beam,

Effective depth of beam (d) = depth of beam – clear cover – diameter/2

$$= 100\text{mm} - 10\text{mm} - 6/2\text{mm}$$

$$= 87\text{mm}$$

We know,

Compression = Tension

$$\text{Or, } 0.36 * \sigma_{ck} * b * x = 0.87 * \sigma_y * A_{st}$$

$$x = \frac{0.87 \times 500 \times 2 \times \frac{\pi}{4} \times 6^2}{0.36 \times 2 \times 100}$$

$$x = 34.16 \text{ mm}$$

Distance of neutral axis (x_m) = $0.46 * d$

$$= 0.46 * 87\text{mm}$$

$$= 40.02 \text{ mm}$$

Here, $x < x_m$ so, it is under reinforced condition.

i. For under reinforced beam with Stirrups

Number of rebars = 4

Diameter of rebars = 6 mm

Clear Spacing = $5 \text{ cm} * 2$

Length of rebar = $50\text{cm} - 5 \text{ cm} * 2$

$$= 40 \text{ cm}$$

Total length of rebars = $4 * 40\text{cm}$

$$= 160 \text{ cm}$$

For stirrups,

Clear spacing on each side = 10 mm

Length of one stirrup = $4 * (100 - 10 * 2)$ mm

= 320mm

For extra bent part,

Length = $2 * 6 * \text{Diameter of stirrup } (\varnothing)$

= $2 * 6 * 6$

= 72mm

Total length of one stirrup = 320mm + 72mm

= 392 mm

Number of stirrups = 5 (Assume)

So, Total length of stirrups = $5 * 392$ mm

= 1960mm

= 1.96m

For spacing of stirrups,

Clear cover at top and bottom = 5cm each

Spacing of stirrups = $\frac{\text{Length of rebar without the clear cover}}{\text{No of stirrups} - 1}$

= $\frac{40 - 5 * 2}{5 - 1}$

= 7.5cm

Table 4. Flexural Strength Calculation of Stirrups (Shear Beam)

Flexural Strength Testing of Molded Flexure Test Specimens IS 516:1959/2021										
Location:	Concrete Technology Lab, KU				Brand of Cement:	Sarbottom				
Date of Casting:	14/05/2024				Aggregate Detail:	20mm passing & 10 mm retaining				
Date of Testing:	11/06/2024				Concrete Grade:	M20				
Age of Beam:	28 Days				W/C Ratio:	0.5				
Curing Condition:	Water Curing				ID Mark:					
S.N.	Beam No.	Dimension of Beam (mm)	Weight of Specimen (Kg.)	Position of Fracture (mm)	Length of Beam (mm)	Failure Load (N)	Flexural Strength (N/mm ²)	Average Strength (N/mm ²)	Failure Mode	Rem .
1	Stirrups (Shear beam)	500*100*100	12.18	45	400	17530	2.366			28 days

6.4.2 Load vs Deflection Curve of Under-reinforced beam with Shear Stirrups

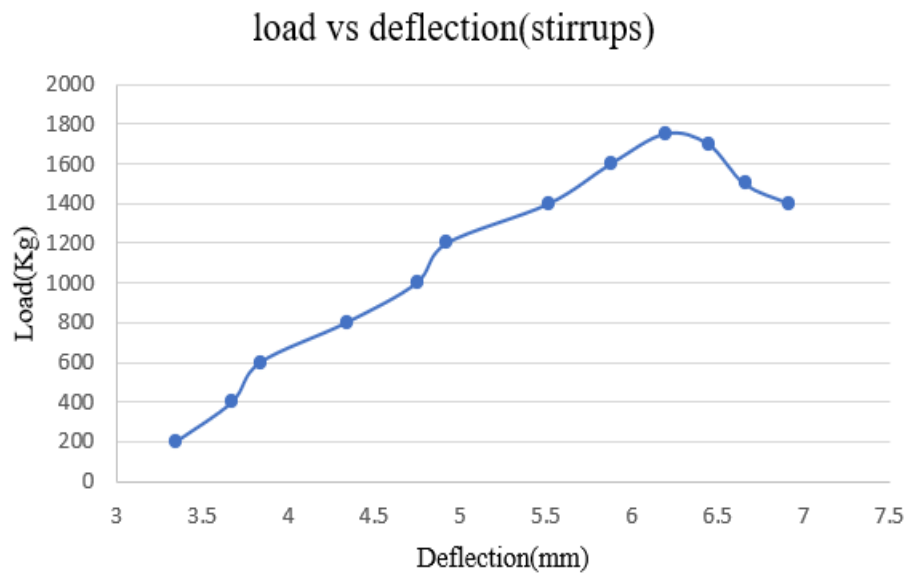


Figure 17. Load vs Deflection Curve of Under-reinforced beam with shear stirrups

From the graph, the ultimate load at which the failure occurs is 1753 kg.

6.5 Combined Load vs Deflection Curve

The comparison of load vs deflection curve among all the cases is shown in **Figure 18**.

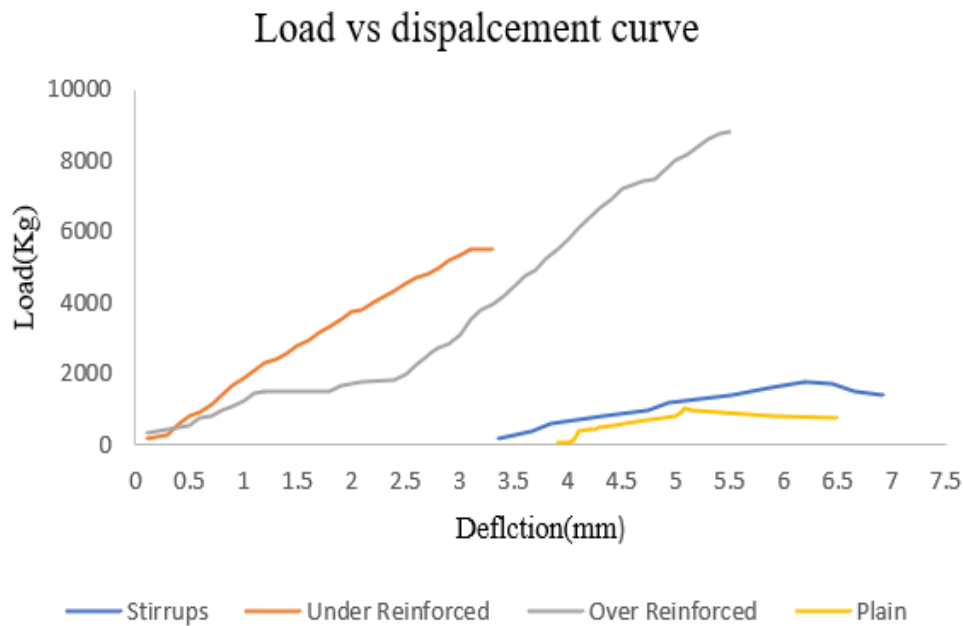


Figure 18. Comparison of Load vs Deflection Curve among all

7 Calculations

7.1 Flexural Strength Calculations

7.1.1 For Plain Beam

Position of Failure (a) = 14.5 cm

As per IS 516.1959 clause 8.4, when a is greater than 13.33 cm for 10 cm specimen then flexural strength (F(b)) is equal to,

$$\begin{aligned}\text{Flexural Strength } (f_b) &= \frac{P L}{b d^2} \\ &= \frac{10320 \times 40}{10 \times 10^2} \\ &= 4.048 \text{ N/mm}^2\end{aligned}$$

The flexural strength of the plain beam is **4.048 N/mm²**

7.1.2 For Over-Reinforced Beam

The flexural strength of the beam is calculated as below:

$$\text{Flexural Strength } (f_b) = \frac{3 P a}{b d^2}$$

Where,

P = maximum load in Kg applied in test specimen

L = span of specimen in cm (60 cm in our case)

B = width of specimen (15 cm in our case)

D = depth of the beam (15 cm in our case)

a = distance of the failure crack from the nearest support

we have from graph,

Failure load (P)= 8827 kg

g = 10 m/s²

Failure load(P) = 8827*10=88270 N

Now,

$$\begin{aligned}
 \text{Flexural Strength } (f_b) &= \frac{3P a}{bd^2} \\
 &= \frac{3 \times 8827 \times 0.1}{15 \times 15^2} \\
 &= 0.7846 \text{ kg/cm}^2 \\
 &= 0.07846 \text{ N/mm}^2
 \end{aligned}$$

∴ Flexural strength (OR) = 0.07846 N/mm²

7.1.3 For Under-Reinforced Beam

The flexural strength of under reinforced section is calculated below:

The flexural strength of the beam is calculated as below:

$$\text{Flexural Strength } (f_b) = \frac{3P a}{bd^2}$$

Where,

P = maximum load in Kg applied in test specimen

b = width of specimen (15 cm in our case)

d = depth of the beam (15 cm in our case)

a = distance of the failure crack from the nearest support

we have from graph,

Failure load (P) = 5520 kg

g = 10 m/s²

failure load(P) = 5520*10= 55200 N

Now,

$$\begin{aligned}
 \text{Flexural Strength } (f_b) &= \frac{3P a}{bd^2} \\
 &= \frac{3 \times 55200 \times 0.5}{15 \times 15^2} \\
 &= 24.533 \text{ kg/cm}^2 \\
 &= 0.245 \text{ N/mm}^2
 \end{aligned}$$

∴ Flexural strength (UR) = 0.245 N/mm²

7.1.4 For Under-reinforced beam with Shear Stirrups

The flexural strength of under reinforced section is calculated below:

The flexural strength of the beam is calculated as below:

$$\text{Flexural Strength } (f_b) = \frac{3P a}{bd^2}$$

Where,

P = maximum load in Kg applied in test specimen

b = width of specimen (15 cm in our case)

d = depth of the beam (15 cm in our case)

a = distance of the failure crack from the nearest support

we have from graph,

Failure load (P)= 1753 kg

$g = 10 \text{ m/s}^2$

failure load(P) = 1753*10= 17530 N

Now,

$$\begin{aligned}\text{Flexural Strength } (f_b) &= \frac{3P a}{bd^2} \\ &= \frac{3 \times 17530 \times 4.5}{10 \times 10^2} \\ &= 23.665 \text{ kg/cm}^2 \\ &= 2.366 \text{ N/mm}^2\end{aligned}$$

$\therefore$ Flexural strength (UR with shear stirrups) = 2.366 N/mm²

8 Results and Conclusions

i. Results

The following results were obtained from the Flexural Testing of the beam.

Table 5. Results of Flexural Testing of Beams

SN	Beam	Flexural Strength (N/mm ²)
1	Plain	4.048
2	Over-reinforced	0.078
3	Under-reinforced	0.245

ii. Conclusions

The determination of the ultimate load and load versus deflection of a molded under-reinforced beam (flexure specimen) provides several key conclusions. The ultimate load reveals the beam's maximum load-carrying capacity, confirming its structural safety and compliance with design specifications. The load-deflection curve indicates the beam's ductility, with significant deflection before failure suggesting good ductility and structural resilience under overload conditions. The initial slope of the curve reflects the beam's stiffness, with a steeper slope indicating higher stiffness and better resistance to deformation. The load at first crack, or cracking load, provides insights into the tensile strength of the concrete and the performance of the reinforcement. Observing the failure mode, whether flexural or shear, helps understand the structural behavior under load, with flexural failure being preferable due to its warning before collapse. The results also assess the reinforcement's efficiency, guiding optimization of reinforcement design to ensure it is sufficient but not excessive. Overall, these conclusions validate the structural design, ensuring safety and reliability in real-world applications.

9 Limitations

- i. The test doesn't include thorough testing like Fineness test, Consistency Test, LA Abrasion Test, etc. of the individual materials used, such as cement, sand, and aggregate,
- ii. Specimens should be cured in a standard water pool, but deviations from this, such as curing in uncontrolled environments, can lead to inconsistent results.
- iii. Laboratory conditions, including temperature and humidity, were not carefully controlled, as variations can affect the curing process and, consequently, the strength of the concrete.
- iv. The loading rate applied during flexural test was not adhered to standard specifications given by the codes, incorrect loading rates can lead to either an underestimation or overestimation of the concrete's flexural strength.
- v. No compressive test was conducted because the testing machine in laboratory was out of order.

10 References

- *Indian Standard code of practice for general structural use of plain and reinforced concrete, IS 456:2000*
- *Compressive, Flexural and Split Tensile Strength, IS 516: 1959/2021*
- *Design of RCC Structural Elements (Vol-I), S.S. Bhavikatti*

11 Annexes



Figure 19. Failure Pattern in Under-reinforced Beam (Left-Side)



Figure 20. Failure Pattern in Under-reinforced (Right-Side)



Figure 21. Failure Pattern in Under-reinforced (Top)



Figure 22. Failure Pattern in Over-Reinforced (Left Side)



Figure 23. Failure Pattern in Over-Reinforced (Right Side)

1 DETERMINATION THE COMPRESSIVE STRENGTH OF CUBES

2 Title

- i. Determination of compressive strength of the cube specimen.

3 Objective

- i. To determine the compressive strength of concrete cube specimens prepared and to verify the strength requirements as desired in the mix design

4 Theory

4.1 Compressive Strength of Concrete

The compressive strength of concrete is a crucial parameter that defines its capacity to withstand axial compressive loads without failure. This property is typically assessed through a standardized test involving concrete cubes, which are cast and cured under controlled conditions to ensure uniform hydration and strength development. After the curing period, these cubes are placed in a compression testing machine, where a gradually increasing load is applied until the specimen fails. The maximum load at which failure occurs, denoted as P , is recorded. The compressive strength f_c is then calculated using the formula:

$$f_c = \frac{P}{A}$$

where A is the cross-sectional area of the cube. The result is expressed in megapascals (MPa). This calculated value is essential for verifying that the concrete meets the specific design criteria and construction standards. Ensuring adequate compressive strength is vital for the structural integrity, safety, and durability of concrete structures, as it directly affects their ability to bear loads, resist various stressors, and maintain their functional performance over their intended lifespan.

4.2 Relation among Compressive Strength and Flexural Strength

According to the Indian Standards IS 516 and IS 456, there is a recognized relationship between the compressive strength and flexural strength of concrete. While compressive strength measures the ability of concrete to withstand axial loads, flexural strength assesses its capacity to resist bending or flexural stresses. IS 456 provides empirical formulas to estimate the flexural

strength from the compressive strength. For ordinary concrete, the flexural strength f_{cr} can be estimated using the following formula:

$$f_{cr} = 0.7\sqrt{f_c}$$

where f_{cr} is the flexural strength in MPa and f_c is the compressive strength in MPa. This relation indicates that as the compressive strength of concrete increases, its flexural strength also increases, but at a lower rate. The empirical formula is useful in preliminary design and quality control, allowing engineers to estimate flexural strength based on more commonly tested compressive strength values. This relationship underscores the inherent connection between the two properties, both of which are critical for ensuring the structural performance and durability of concrete elements in various applications.

5 Methodology

- Clean the mould and apply oil
- Fill the concrete in the mould in layers approximately 5cm thick
- Compact each layer with not less than 35 strokes per layer using a tamping rod (steel bar 16mm diameter and 60cm long, bullet pointed at lower end)
- Level the top surface and smoothen it with a trowel
- The test specimens are stored in moist air for 24 hours and after this period the specimens are marked and removed from the mould and kept submerged in clear fresh water until taken out prior to test. The water for curing should be tested every 7 days and the temperature of water must be at $27 \pm 20^\circ\text{C}$.
- Remove the specimen from water after specified curing time and wipe out excess water from the surface.
- Clean the bearing surface of the testing machine
- Place the specimen in the machine in such a manner that the load shall be applied to the opposite sides of the cube cast. Align the specimen centrally on the base plate of the machine.
- Rotate the movable portion gently by hand so that it touches the top surface of the specimen.
- Apply the load gradually without shock and continuously at the rate of $140\text{kg/cm}^2/\text{minute}$ till the specimen fails
- Record the maximum load and note any unusual features in the type of failure.

6 Observations

Table 6. Compressive Strength Calculations of Cube

Compressive strength of Test Specimens IS 516:1959/2021			
Location:	KU Lab	Brand of Cement:	Sarbottam
Date of Casting:	14/05/2024	Aggregate Detail:	20mm passing
Date of Testing:	11/06/2024	Concrete Grade:	M20
Age of Beam:	28 Days	W/C Ratio:	0.5
Curing Condition:	Water Curing	ID Mark:	

S.N.	Designation of Specimen	Weight of Specimen (kg.)	Loading Face Dimension		Failure Load (KN)	Compressive Strength (N/mm ²)	Average Flexural Strength (N/mm ²)	Detail of Failure Mode
			Length (mm)	Breadth (mm)				
1	1	2.18	100	100	254	25.4	24.03	Compressive
2	2	2.265	100	100	226.6	22.66		Compressive

7 Calculations

We know compressive strength (f_c) = $\frac{\text{Failure Load (N)}}{\text{Area of Face on which load applied (mm}^2\text{)}}$

$$\text{Cube No 1: } f_c = \frac{254000}{100 \times 100} = 25.4 \text{ N/mm}^2$$

$$\text{Cube No 2: } f_c = \frac{226600}{100 \times 100} = 22.66 \text{ N/mm}^2$$

$$\text{Cube No 3: } f_c = \frac{240300}{100 \times 100} = 24.03 \text{ N/mm}^2$$

$$\text{Average} = 24.03 \text{ N/mm}^2$$

8 Results and Conclusions

8.1 Results

Hence the average compressive strength of cube was found to be 24.03N/mm^2

8.2 Conclusions

Compressive strength of M20 concrete in 28 days should be 20 N/mm^2 and our observed value of compressive strength is 24.03 N/mm^2 which is satisfactory. Hence, the concrete cube we design for M20 was good.

9 Limitations

- Test of materials like cement, sand, aggregate was not performed.
- In case of steel reinforcement, due to rusting the actual diameter of steel may be reduced.
- Load provided to specimen was in short period of time but in actual case load is acted on beam over long period of time (creep).
- The curing tank temperature was not maintained the same throughout curing period.
- Immediate curing after 24 hours was not done which is responsible for decrease in strength of specimen.

10 References

- *Indian Standard code of practice for general structural use of plain and reinforced concrete, IS 456:2000*
- *Compressive, Flexural and Split Tensile Strength, IS 516: 1959/2021*
- *Design of RCC Structural Elements (Vol-I), S.S. Bhavikatti*

11 Annexes



Figure 24. Cube Specimen